

PRAKTIKUM BASIS DATA

MODUL 09

JURNAL



Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

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JURNAL

1. Soal

Running [code sql](#) berikut, screenshot hasilnya dan analisis hasil code tersebut

1. create table employees as select * from hremployees.
2. select * from employees.
- 3.

```
1 CREATE OR REPLACE PROCEDURE increase_salary (  
2 department_id_in IN employees.department_id%TYPE,  
3 increase_pct_in IN NUMBER)  
4 IS  
5 BEGIN  
6     FOR employee_rec  
7         IN (SELECT employee_id  
8             FROM employees  
9             WHERE department_id = increase_salary.department_id_in)  
10    LOOP  
11        UPDATE employees emp  
12            SET emp.salary = emp.salary + emp.salary * increase_salary.increase_pct_in  
13            WHERE emp.employee_id = employee_rec.employee_id;  
14        DBMS_OUTPUT.PUT_LINE ('Updated ' || SQL%ROWCOUNT);  
15    END LOOP;  
16 END increase_salary;  
17
```

4. BEGIN
 increase_salary (90, .50);
 ROLLBACK;
END;
/
select * from employees where department_id = 90
5. Apa yang terjadi jika statement ROLLBACK dihapus pada sql code no 4.

Jawaban:

TABEL

```

SET SERVEROUTPUT ON

CREATE TABLE hemployees (
  employee_id NUMBER(6) NOT NULL,
  first_name VARCHAR2(20),
  last_name VARCHAR2(25) NOT NULL,
  email VARCHAR2(25) NOT NULL,
  phone_number VARCHAR2(20),
  hire_date DATE NOT NULL,
  job_id VARCHAR2(10) NOT NULL,
  salary NUMBER(8,2),
  commission_pct NUMBER(2,2),
  manager_id NUMBER(6),
  department_id NUMBER(4),
  CONSTRAINT emp_emp_id_pk PRIMARY KEY (employee_id)
);

-- INSERT data untuk department_id = 90 (Executive — dipakai soal no.4)
INSERT INTO hemployees VALUES (100, 'Steven', 'King', 'SKING', '515.123.4567', TO_DATE('17-JUN-2003','DD-MON-YYYY'), 'AD_PRES', 24000, NULL, NULL, 90);
INSERT INTO hemployees VALUES (101, 'Neena', 'Kochhar', 'NKOCHHAR', '515.123.4568', TO_DATE('21-SEP-2005','DD-MON-YYYY'), 'AD_VP', 17000, NULL, 100, 90);
INSERT INTO hemployees VALUES (102, 'Lex', 'De Haan', 'LDEHAAN', '515.123.4569', TO_DATE('13-JAN-2001','DD-MON-YYYY'), 'AD_VP', 17000, NULL, 100, 90);

-- INSERT tambahan department lain (agar SELECT * tidak kosong)
INSERT INTO hemployees VALUES (103, 'Alexander', 'Hunold', 'AHUNOLD', '590.423.4567', TO_DATE('03-JAN-2006','DD-MON-YYYY'), 'IT_PROG', 9000, NULL, 102, 60);
INSERT INTO hemployees VALUES (104, 'Bruce', 'Ernst', 'BERNST', '590.423.4568', TO_DATE('21-MAY-2007','DD-MON-YYYY'), 'IT_PROG', 6000, NULL, 103, 60);
INSERT INTO hemployees VALUES (107, 'Diana', 'Lorentz', 'DLORENTZ', '590.423.5567', TO_DATE('07-FEB-2007','DD-MON-YYYY'), 'IT_PROG', 4200, NULL, 103, 60);
INSERT INTO hemployees VALUES (124, 'Kevin', 'Mourgos', 'RMOURGOS', '650.123.5234', TO_DATE('16-NOV-2007','DD-MON-YYYY'), 'ST_MAN', 5800, NULL, 100, 50);
INSERT INTO hemployees VALUES (141, 'Trenna', 'Rajs', 'TRAJS', '650.121.8009', TO_DATE('17-OCT-2003','DD-MON-YYYY'), 'ST_CLERK', 3500, NULL, 124, 50);
INSERT INTO hemployees VALUES (149, 'Eleni', 'Zlotkey', 'EZLOTKEY', '011.44.1344.429018', TO_DATE('29-JAN-2008','DD-MON-YYYY'), 'SA_MAN', 10500, .2, 100, 80);
INSERT INTO hemployees VALUES (174, 'Ellen', 'Abel', 'EABEL', '011.44.1644.429267', TO_DATE('11-MAY-2004','DD-MON-YYYY'), 'SA_REP', 11000, .3, 149, 80);

```

Script Output x

Task completed in 0.127 seconds

Table HREMPLOYEES created.

1 row inserted.

1 row inserted.

1 row inserted.

1.

```

DROP TABLE employees;
CREATE TABLE employees AS SELECT * FROM hemployees;

```

Script Output x

Task completed in 0.057 seconds

Table EMPLOYEES dropped.

Table EMPLOYEES created.

2.

```

SELECT * FROM employees;

```

Query Result x

SQL | All Rows Fetched: 10 in 0.047 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	100	Steven	King	SKING	515.123.4567	17-JUN-03	AD_PRES	24000	(null)	(null)	90
2	101	Neena	Kochhar	NKOCHHAR	515.123.4568	21-SEP-05	AD_VP	17000	(null)	100	90
3	102	Lex	De Haan	LDEHAAN	515.123.4569	13-JAN-01	AD_VP	17000	(null)	100	90
4	103	Alexander	Hunold	AHUNOLD	590.423.4567	03-JAN-06	IT_PROG	9000	(null)	102	60
5	104	Bruce	Ernst	BERNST	590.423.4568	21-MAY-07	IT_PROG	6000	(null)	103	60
6	107	Diana	Lorentz	DLORENTZ	590.423.5567	07-FEB-07	IT_PROG	4200	(null)	103	60
7	124	Kevin	Mourgos	RMOURGOS	650.123.5234	16-NOV-07	ST_MAN	5800	(null)	100	50
8	141	Trenna	Rajs	TRAJS	650.121.8009	17-OCT-03	ST_CLERK	3500	(null)	124	50
9	149	Eleni	Zlotkey	EZLOTKEY	011.44.1344.429018	29-JAN-08	SA_MAN	10500	0.2	100	80
10	174	Ellen	Abel	EABEL	011.44.1644.429267	11-MAY-04	SA_REP	11000	0.3	149	80

3.

```

CREATE OR REPLACE PROCEDURE increase_salary (
    department_id_in IN employees.department_id%TYPE,
    increase_pct_in IN NUMBER)
IS
BEGIN
    FOR employee_rec
    IN (SELECT employee_id
        FROM employees
        WHERE department_id = increase_salary.department_id_in)
    LOOP
        UPDATE employees emp
        SET emp.salary = emp.salary + emp.salary * increase_salary.increase_pct_in
        WHERE emp.employee_id = employee_rec.employee_id;

        DBMS_OUTPUT.PUT_LINE ('Updated ' || SQL%ROWCOUNT);
    END LOOP;
END increase_salary;
/

```

Script Output x

Task completed in 0.482 seconds

Procedure INCREASE_SALARY compiled

4.

```

BEGIN
    increase_salary (90, .50);
    ROLLBACK;
END;
/

```

Script Output x

Query Result x

All Rows Fetched: 3 in 0.014 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	100	Steven	King	SKING	515.123.4567	17-JUN-03	AD_PRES	24000	(null)	(null)	90
2	101	Neena	Kochhar	NKOCHHAR	515.123.4568	21-SEP-05	AD_VP	17000	(null)	100	90
3	102	Lex	De Haan	LDEHAAN	515.123.4569	13-JAN-01	AD_VP	17000	(null)	100	90

5. Jika pernyataan ROLLBACK dihapus, maka perubahan data berupa kenaikan gaji sebesar 50% tidak akan dibatalkan. Perubahan tersebut tetap tersimpan dalam sesi database dan menunggu proses commit. Akibatnya, saat perintah `SELECT * FROM employees WHERE department_id = 90` dijalankan, nilai pada kolom *salary* untuk pegawai di departemen 90 akan terlihat sudah meningkat sebesar 50% dari nilai awalnya.